

Timeline:

1. Compute change features to go alongside every regular feature
 - a. Absolute change (delta)
 - b. Percentage change
2. Suppress Leakable features (median income, home value, rent, college attainment)
 - a. Kept static features, and removed delta and pct versions
3. Exclude Zillow home value data (Used to construct gentrification labels)
4. Add city dummy variables so we can differentiate between different housing markets
5. LASSO model is used for feature selection
 - a. Removes redundant features
6. Tried Logistic Regression, Random Forest, and XGBoost on the LASSO feature set
 - a. Used scale_pos_weight to compensate for the imbalance of positive/negatives in the dataset
7. Ensemble of XGBoost, LR, and RF to provide risk scores
 - a. Ensemble the probability scores generated by each model, averaged together by tract to generate the most reliable risk score, since each model does better with different relationships between features.
 - i. XGBoost: Best for nonlinear interactions
 - ii. Random Forest: Robust to outliers
 - iii. Linear Regression: Stable linear baseline
8. XGBoost is the strongest individual predictor